

Simple Calculation

Assumptions:
1 GB file = $1 \times 1024 \times 1024 \times 1024$ bytes $\approx 1,073,741,824$ bytes
1 character ≈ 1 byte
1 token ≈ 4 characters (typical for English text)
Total tokens $\approx 1,073,741,824 / 4$ $\approx 268,435,456$ tokens ≈ 268 million tokens
1GB File = ≈ 268 million tokens
Choosing chunking parameters
Common ingestion settings: chunk_size = 500 tokens chunk_overlap = 50 tokens
step = chunk_size - chunk_overlap = 500 - 50 = 450 tokens
Number of chunks $\approx$ Total tokens / step $\approx 268,435,456 / 450$ $\approx 596,523$ chunks
Embedding Size Assume:
Embedding dimension = 1536 Float32 per number = 4 bytes $1536 \times 4 = 6144$ bytes ≈ 6 KB (per embedding)
Total Embedding size:
$596,523 \times 6$ KB $\approx 3,579,138$ KB $\approx 3,493$ MB ≈ 3.41 GB
Note: Just for embeddings. Metadata and vector index overhead will increase total size.
Retrieval & Context Window Example Assume LLM context window = 8K tokens
Retrieve top 10 chunks $\times 500$ tokens each = 5,000 tokens
Add user prompt 200 tokens + system prompt 800 tokens + response 1,000 tokens Total $\approx 7,000$ tokens $< 8,000$ token window